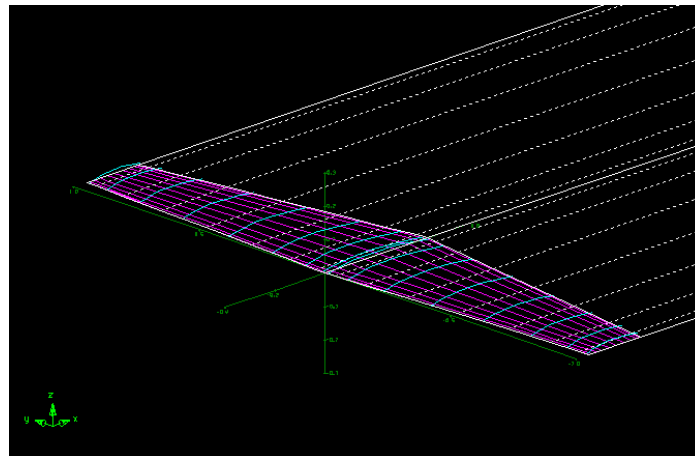




Flying Liftercraft

Wing planform optimisation and propeller interaction analysis



Daniel Earl
CID: 01730804

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Subteam	Members
Technical Directors	Foster Ng
	Julian Fung
	Daniel Earl
Aerodynamics	Anthony Haiser
	Kwok Leung Yim
	Marius Borsu Lefevre
Structures	Sorawis Itthithavorn
	Antoine Rieuneau
	Olivier Sussfeld
	Jordi Frittoli
Propulsion and Systems	Matan Hamburger
	Wei Han Lee
	Natalia Melo Fernandez
	Farsai Sittichoksuwan
	Kristoff Ahlner
Performance, Stability & Control	Joshua Binns
	Elia Chimonides
	Luis Ferreira da Silva Marques
	Yi En Puah
	Bojun Vito Yang
Simulation	Gabisanth Seyon
	Rohit Singh

Abstract

The conceptual design of a UAV wing was iteratively developed in conjunction with other disciplines to abide by constraints of payload, stall speed and wing span. AVL was used as the primary tool for determining lift, lift distribution, maximum lift coefficient and stall location. It was programmed through the command line using Xfoil data to estimate the viscous drag by integrating along the span. The large diameter vertical propellers which provided VTOL capability for the aircraft were analysed using the VSPAERO vortex lattice method. The wing's lift was reduced by the propeller wake but this was most significant for low-mounted propellers hence driving the design decision towards a high-positioning of the propellers.

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1 Introduction

Every year there are many tragic incidents of migrant deaths at sea. The large majority of these deaths are due to drowning because rescue vessels and coastguard personnel do not reach the stranded people in time. To reduce the time taken between reports of a capsized boat and aid arriving a conceptual design into an Unmanned Aerial Vehicle (UAV) has been proposed. With the ability to fly 2.5 times faster than the top speed of common coastguard boats this proposal of a UAV will be able to significantly increase the survival rates of migrants who are endangered.

To design a successful UAV the wing needed to be designed such that it could operate in a large flight envelope that involved a high cruise speed and a low stall speed while minimising drag in order to save fuel. Given the tight time constraints on the project and the pressure from other teams for an initial wing design, an accurate and quick tool was needed to optimise the wing planform. Initially a lifting line theory was used but this was superseded by the superior Athena Vortex Lattice method which drove the majority of design decisions. Later on in the project STAR CCM simulations were run. Given the time required to set up an accurate (Reynolds Averaged Navier-Stokes) RANS simulation, these analyses lagged behind the iterative design process and could not be used to drive decision-making. However, the STAR CCM simulations were helpful in applying corrections to the AVL method and validating the accuracy of the wing design. This report focuses on how AVL was used to optimise the wing planform in conjunction with the other design disciplines.

2 Vortex Lattice Method theory

The Vortex Lattice Method (VLM) which is used in the Athena Vortex Lattice (AVL) software [2] is an inviscid and incompressible solver which assumes steady flow. These assumptions mean that velocity $V = \nabla\phi$ for some function ϕ . Applying these assumptions to the Navier-Stokes equations leads to a nonlinear partial differential equation (PDE) known as the velocity potential equation [3]. However, making the further assumption of small velocity perturbations, a linear PDE governed by Laplace's equation emerges as follows

$$(1 - M_\infty^2) \frac{\partial^2 \hat{\phi}}{\partial x^2} + \frac{\partial^2 \hat{\phi}}{\partial y^2} = 0 \quad (1)$$

where M_∞ is the freestream Mach number and $\hat{\phi}$ is the perturbation velocity potential.

Since equation 1 is linear, superpositions of elementary solutions to it yield more complicated flow regimes which are still solutions to this equation. One example of an elementary solution to this is the vortex line (or filament). The idea behind the vortex lattice method is to seek to position horseshoe vortex lines in such a way as to model the flow about a body such as a wing. In Prandtl's lifting line theory an infinite number of horseshoe vortices are placed along a single lifting line. However, in vortex lattice methods the horseshoe vortices are placed in both the spanwise and chordwise directions as shown in figure 1

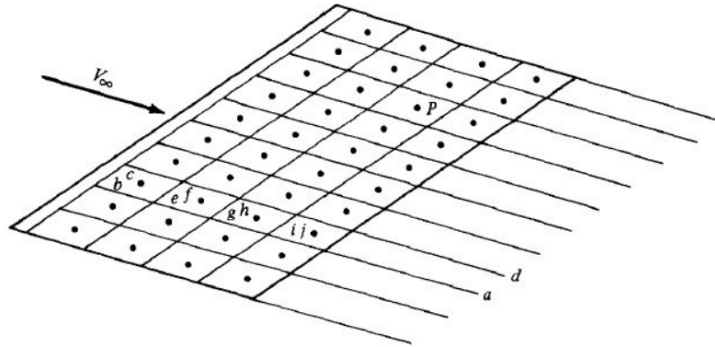


Figure 1: Vortex lattice layout on wing [1, p. 473]

The Biot-Savart law can be used to calculate the velocity at any point i induced by a vortex line. Hence the perturbation velocity $\nabla\phi_i$ induced at a single collocation point by all the vortex lines is given by

$$\nabla\phi_i = \sum_{j=1}^N w_{ij}\Gamma_j \quad (2)$$

w_{ij} is known as the influence matrix containing the geometry of the locations of the vortex lines and Γ_j is the strength of each vortex. The flow at each collocation point must be parallel to the camber line. Therefore, by denoting the normal vector at each collocation point i as n_i we require that $v_i \cdot n_i = 0$. This can be written in terms of the influence matrix as

$$(V_\infty + \sum_{j=1}^N w_{ij}\Gamma_j) \cdot n_i = 0 \quad (3)$$

This system of equations is solved for all Γ_j to find the strength of each horseshoe vortex situated around the body.

The Kutta-Joukowski theorem relates lift to circulation strength Γ . Since the vortex strength at each lattice position on the body is known the lift at each spanwise position can be determined. The drag can be calculated from integrating the pressure around the aerofoil or using an alternative method that analyses far-field flow on what is called the Trefftz plane [4]. The method that AVL uses for finding the induced drag is by using a Trefftz plane analysis - the derivation of which can be found at [5].

The advantage of VLM methods over the lifting line method is that it captures the effect of camber. However, it does not account for the thickness of the aerofoil. Though at this conceptual design stage it is reasonable to neglect aerofoil thickness especially given that the aerofoils considered are relatively thin.

3 Initial wing sizing and aerofoil selection

The initial design of the wing was driven by the mission profile that the UAV was required to fulfil. The furthest range that the UAV needed to travel to get to a capsized migrant boat was set at 70 km. In order to arrive at the incident significantly before coastguard ships can, this distance needed to be covered in 20 minutes. Hence the cruise speed was 210 km/h = 58.3 m/s. As an initial estimate, the stall speed V_s was set at roughly 70% of the cruise speed yielding $V_s=40$ m/s. This was the assumed stall speed for multiple iterations of the wing design until the Control team could specify in greater detail how the transition from VTOL to forward flight needed to occur. From initial estimates of the layout of the UAV the maximum take off weight (MTOW) was predicted to be 28.4 kg. A density of $\rho=1.21$ at the cruise height of 400 ft was used and it was assumed that the lift required in the cruise needs to balance the MTOW. The equation for lift L is given by

$$L = \frac{1}{2}\rho V^2 S_{ref} C_L \quad (4)$$

Using $V = V_s$ and $C_L = C_{Lmax}$ (the maximum lift coefficient of the wing) means that there are still two free variables: namely the wing reference area S_{ref} and C_{Lmax} . Initially, C_{Lmax} was assumed to be 1.1. Rearranging equation 4 results in $S_{ref}=0.26$ m². This area was the starting point for the wing design iterative process which allowed other disciplines such as Control to start working on stability analysis.

Mach number is defined as $M = \frac{V_\infty}{\sqrt{\gamma RT}}$ where γ is the ratio of specific heats, R is the gas constant and T is the temperature. Hence for our UAV the Mach number at the cruise is $M = 58.3/(\sqrt{1.4 \cdot 287 \cdot 287.4}) = 0.17$. This means that the cruise velocity is significantly less than 30% of the speed of sound meaning that, as a first approximation, compressibility effects can be ignored. For this reason the wing was given a sweep of 0° at the quarter chord position.

Next an aerofoil needed to be decided upon. The aim was to choose an aerofoil that has a low lift to drag ratio (L/D) at the design lift coefficient of the cruise. Since the start of the cruise is soon after the vertical take off it was assumed that the lift in the cruise needed to balance the MTOW. Applying equation 4 at the cruise speed of 58.3 m/s and using $S_{ref}=0.26$ m² gave a cruise $C_L = 0.51$. However, to take into account 3D effects, the 2D C_L of the aerofoil was assumed to be 0.60 (17% higher than the 3D wing's C_L). Aerofoils are designed to operate best within a range of Reynolds numbers (Re). Assuming an aspect ratio $AR = 7$ and a taper $\lambda = 1$ results in a span b of 1.36 m and a mean aerodynamic chord (MAC) of 0.19 m. Hence $Re = 1.21 \cdot 58.3 \cdot 0.19 / (1.8 \times 10^{-5}) = 7.4 \times 10^5$.

In order to assess the characteristics of 2D aerofoils the software Xfoil [6] may be used. In the interest of providing a quick turnaround in the initial wing design so that other disciplines in the group could start their analysis it was decided to use the popular website Airfoil Tools [7] instead of trying to analyse many aerofoils directly using the Xfoil software. It provides aerofoil data at a range of Re which saves having to run Xfoil multiple times. The downside is that our Re lay in between the two values of Re that Airfoil Tools provided. A Python script running the requests [8] library was used to access data from over 110 aerofoils from the NACA 4, 5 and 6 families. Supercritical aerofoils were not considered given the low Mach number at the cruise. For each aerofoil the L/D was calculated at the 2D design C_L of 0.60. The aerofoil's C_{Lmax} was also found at $C_L = 0.60$ in order to assess whether the aerofoil could comfortably give a C_L of $1.1 \times 1.17 = 1.29$ (our assumed C_{Lmax} with the 17% correction factor applied) without being near the stall. The NACA 5 family of aerofoils with the highest L/D ratios had lower ratios than the best aerofoils in the NACA 4 and 6 families. So the top three aerofoils in terms of L/D from the NACA 4 and 6 families were shortlisted. The aerofoil data evaluated at the 2D c_l of 0.60 is given in table 1.

Table 1: L/D and C_{Lmax} for select aerofoils

Aerofoil name	$Re = 10^6$		$Re = 5 \times 10^5$	
	L/D	C_{Lmax}	L/D	C_{Lmax}
NACA 2410	97.06	1.53	82.48	1.36
NACA 2411	100.10	1.42	85.16	1.30
NACA 4412	101.96	1.67	85.15	1.52
NACA 64a410	98.34	1.34	95.78	1.24
NACA 651412	103.68	1.34	78.15	1.24
NACA 65410	110.53	1.30	87.11	1.16

Despite the 65410 aerofoil having the largest L/D ratio it also has the lowest C_{Lmax} . Even at this early stage the expectation was that the taper ratio of the wing would be lower than 1 in order to more closely resemble an elliptical distribution. So Re near the tip would be less than the average value and hence would be close to $Re = 5 \times 10^5$. In order to prevent the tip from stalling, the 2D C_{Lmax} at $Re = 5 \times 10^5$ needed to be greater than the previously determined value of 1.29. None of the shortlisted NACA 6 aerofoils met this condition. From the three remaining NACA 4 aerofoils the 2411 was ruled out given that its C_{Lmax} at $Re = 5 \times 10^5$ was less than 1% higher than necessary to prevent stall.

Leung [9] conducted an initial estimate of the ideal planform layout for the 2410 aerofoil using lifting line theory with 6 collocation points. To build on this method the Athena Vortex Lattice software was run through the command line. As explained in section 2, AVL is able to model the curvature of the aerofoil whereas lifting line theory is unable to making AVL more accurate. In both the lifting line theory and AVL analysis a value for the induced drag coefficient (C_{Di}) was calculated for different taper ratios (λ) and twist (θ). However, when running AVL through the command line, an additional functionality was implemented to calculate an estimate of the viscous drag (C_{Dv}) comprising of pressure drag and skin friction drag. This was done by incorporating 2D Xfoil analysis and integrating spanwise as explained in procedure 1.

Procedure 1.

- Run Xfoil for the test aerofoil for a range of Reynolds numbers $4 \times 10^4 \leq Re \leq 1.5 \times 10^7$ and angles of attack $-10^\circ \leq \alpha \leq 20^\circ$ and save the C_{Dv} for each Re - α pair in a matrix.
- Linearly interpolate to find the local chord at each spanwise position and hence the local Re along the span.
- Run AVL at the α that gives the 3D cruise $C_L = 0.51$.
- AVL gives C_{Di} using a Trefftz plane analysis.
- AVL gives the induced angle of attack (α_i) at each strip of the wing.
- Hence calculate the local effective angle of attack (α_{eff}) at each spanwise position along the wing using 5 which includes the effect of local twist (θ_i).

$$\alpha_{eff} = \alpha + \theta_i - \alpha_i \quad (5)$$

- At each spanwise position, use the $Re-\alpha$ to interpolate the Xfoil matrix to give a local C_{Dv} along the span.
- Integrate the local C_{Dv} and add this to AVL's C_{Di} to provide a total drag coefficient C_D .

To verify the drag calculation procedure the software Xflr5 [10] was run using a test case with the NACA 4412 aerofoil, $\lambda = 0.55$ and $\theta = -4^\circ$. Xflr5 uses the same method as was used in the drag calculation procedure explained above. Table 2 shows that the two methods provide similar results suggesting that the integration of the 2D C_{Dv} data had been performed correctly. The reason for the discrepancy in C_{Di} between the two methods is likely due to the fact that the inviscid solver in Xflr5 was a panel method. This accounts for the thickness of the aerofoil whereas AVL only accounts for the camber. However, the benefit of finding the total drag alongside running AVL is that multiple λ - θ parameters can be tested in a loop whereas Xflr5 requires manual user-input since it is only GUI-based.

Table 2: Comparison of Matlab and Xflr5 drag prediction

	C_{Di}	C_{Dv}
Xflr5	0.0114	0.0070
Matlab	0.0123	0.0069
% difference	8.4	1.4

At this stage S_{ref} was 0.26 m^2 and it was still assumed that $AR = 7$. Aspect ratio is defined as

$$AR = \frac{b^2}{S_{ref}} \quad (6)$$

which can be rearranged to give span $b = 1.35 \text{ m}$. Simple geometry dictates that the root chord is given by

$$c_r = \frac{2S_{ref}}{b(1 + \lambda)} \quad (7)$$

while the tip chord is

$$c_t = \lambda c_r \quad (8)$$

The above stated planform parameters were used and equations 7 and 8 solved for each of the NACA 2410 and 4412 aerofoils in the range $0.2 \leq \lambda \leq 1$ in steps of 0.05 and $-4^\circ \leq \theta \leq 0^\circ$ in steps of 0.25. Then the drag calculation procedure outlined above was conducted on each aerofoil for each combination of λ - θ values. Surfplots of the variation of C_D with λ and θ is shown in figure 2.

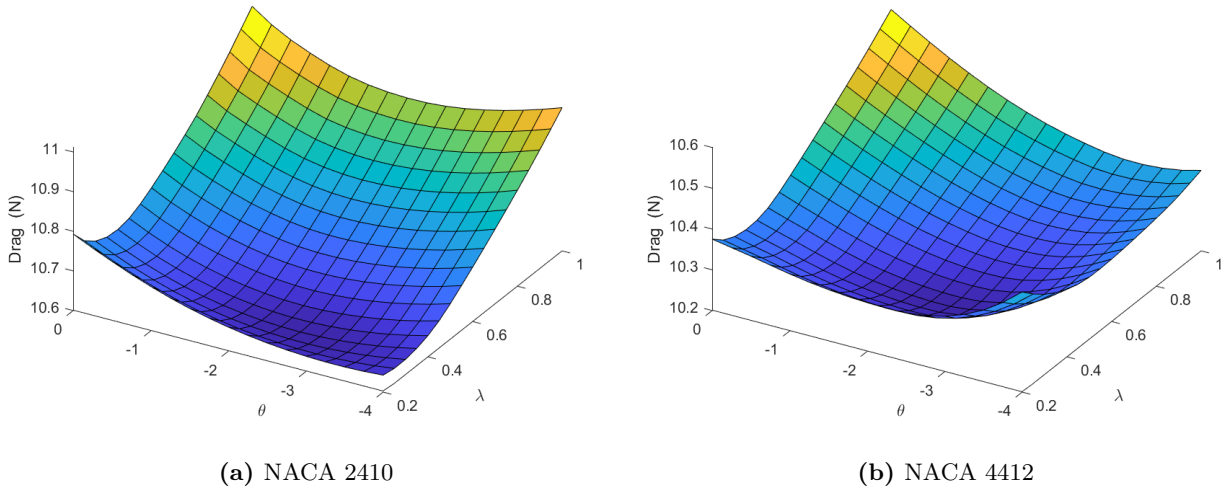


Figure 2: Drag vs λ and θ

For each aerofoil there was a λ - θ pair that minimised the drag. These minima are detailed in table 3.

Table 3: Minimum drag planforms for NACA 2410 and 4412

Aerofoil	Drag (N)	λ	θ°
2410	10.61	0.35	-2.5
4412	10.25	0.45	-2.5

Since the NACA 4412 aerofoil gave a slightly lower drag than the 2410 aerofoil it was chosen as the final design. Instead of the optimum taper of 0.45 for this aerofoil the Control team requested that λ be set at 0.5 because there was a lot of experimental data for determining stability derivatives at this value but not at 0.45. For the initial choice of wing planform this compromise was deemed suitable.

At this stage the wing was optimised to produce minimum drag at the cruise while providing a stall speed of 40 m/s. However, this was assuming that the wing could achieve a C_{Lmax} of 1.1. The next section outlines the process of estimating the C_{Lmax} of the wing to see how it compared to the assumed value of 1.1.

4 Estimating C_{Lmax}

From the outset it was assumed that the onset of stall occurred when one part of the wing had just reached a stalled condition. Just as Xfoil can be used to give a sectional viscous drag coefficient, it can also give the sectional C_L at which stall occurs. Following a similar method for integrating sectional viscous drag, the way that the wing's C_{Lmax} was calculated was as follows:

Procedure 2.

- Run Xfoil for the test aerofoil for a range of Reynolds numbers $4 \times 10^4 \leq Re \leq 1.5 \times 10^7$ and angles of attack $-10^\circ \leq \alpha \leq 20^\circ$ and save the sectional c_{lmax} for each Re in a matrix.
- Linearly interpolate to find the local chord at each spanwise position and hence the local Re along the span.
- Hence find the local c_{lmax} at each spanwise position.
- Run AVL at the required stall speed and (initially) an arbitrary α .
- AVL gives the local lift coefficient (c_l) at each spanwise position.
- If $c_l > c_{lmax}$ at more than spanwise position then re-run AVL with a lower α .
- If $c_l < c_{lmax}$ for all spanwise positions then re-run AVL with a higher α .
- If $c_l \approx c_{lmax}$ at only one spanwise position (to within a precision of 0.01) then the current α stalls the wing.
 - The total lift coefficient (C_L) at this α is the wing's C_{Lmax} .
 - The spanwise location at which $c_l \approx c_{lmax}$ is the position at which stall first occurs on the wing.

Using the C_{Lmax} procedure it was found that the actual C_{Lmax} of the wing was 1.40. This is higher than the assumed value of 1.1. Hence S_{ref} did not need to be as high. To find the wing area that provided a $V_s=40$ m/s it was not as simple as correcting C_{Lmax} to 1.40 and rearranging equation 4 for S_{ref} . This is because C_{Lmax} is a function of the wing's area. This dependence of C_{Lmax} on S_{ref} is due to the chord lengths (and hence Re) reducing when S_{ref} is reduced (assuming a fixed AR). And since the 2D aerofoil's c_{lmax} depends on Re the 3D C_{Lmax} depends on Re . For this reason, an iterative loop was set up as detailed in procedure 3.

Procedure 3.

1. Choose an (initially) arbitrary C_{Lmax} .
2. Rearrange equation 4 (evaluated at the stall) for S_{ref} .
3. Use equations 6, 7 and 8 to determine the other planform parameters.
4. Do procedure to find this wing's actual C_{Lmax} .

5. Use this actual C_{Lmax} in step 1 and repeat steps 2-4 until the start and end C_{Lmax} values have converged.

Performing procedure 3 meant S_{ref} reduced from the initial design of 0.26 m^2 down to 0.21 m^2 with an associated $C_{Lmax} = 1.37$.

From this point onwards the estimation of C_{Lmax} was done using AVL and procedure 2 but the software Xflr5 was used to find both C_D and C_L as a function of α . The reason for this was that Xflr5 was found to more quickly analyse a range of α values compared to using procedure 1 to repeatedly write files to disk by running AVL through the command line. Having corrected the wing planform to account for the actual C_{Lmax} achievable by the wing, the C_L at the cruise was determined. This was done by rearranging equation 4 for C_L using $S_{ref} = 0.21 \text{ m}^2$ and $V = 58.3 \text{ m/s}$ (the cruise speed) yielding $C_L = 0.65$. This C_L was achieved at $\alpha = 4.71^\circ$ at which point $C_D = 0.0253$. Hence the wing's drag at the cruise was 10.9 N .

In summary, since the wing could out-perform the initial assumption of $C_{Lmax} = 1.1$ this resulted in the wing planform area reducing while still meeting the $V_s=40 \text{ m/s}$ requirement. Having optimised the wing aerodynamically it was time to perform a multi-disciplinary optimisation process with the other Functional Area Teams (FATs). This is detailed in the next section.

5 Multi-disciplinary optimisation

5.1 Camera frame rate requirement on stall speed

The most urgent issue that needed to be addressed at this stage with the wing design was the high stall speed of 40 m/s . The Control team were still finalising the stall requirements they needed to perform a successful transition from vertical to forward flight. However, it was realised that the 40 m/s stall speed did not even satisfy the frame rate requirements of the camera which needs to identify the migrants.

Marques [11] calculated that in order to fly slow enough for the camera to capture enough frames of each area flown over, the loiter speed used for searching should be 40 m/s . Initially a safety factor of 1.1 was applied to this to calculate the new requirement for V_s . However Ng [12] had performed an analysis on the regulation requirements for the UAV and found that it needed to withstand forward or aft gusts of 7.7 m/s . This was included in the new V_s calculation to arrive at $V_s = \frac{40-7.7}{1.1} = 29.4 \text{ m/s}$.

Further to the new stall speed constraint, the initial assumption of $AR = 7$ was superseded by a constraint on the span b . Ng found that the smallest type of water vessel our UAV could be expected to operate from had a deck size of $3 \text{ by } 1.5 \text{ m}$. Hence b was fixed at 2 m to allow for 0.5 m clearance at each wing tip.

The new stall speed of 29.4 m/s was input to procedure 2 along with the requirement that $b = 2 \text{ m}$ and the new wing planform changed significantly. Given the larger operating envelope that the wing had to satisfy (58.3 to 29.4 m/s compared with 58.3 to 40 m/s before), S_{ref} increased significantly.

However, with a larger wing, the α required to cruise was reduced to 0.52° . This reduction in cruise α meant that C_D at cruise reduced to 0.0116 . C_D reduced further in this iteration than the increase in S_{ref} so the total drag at the cruise reduced to 9.3 N . In changing the initial constraint of $AR = 7$ to $b = 2$ the AR of this new iteration increased to 10.3 . This increase in AR reduced the induced drag and hence increased the aerodynamic efficiency of the wing. This meant that it could achieve lower stall speeds than the iteration in section 4 while also encountering a reduction in the drag at the cruise.

5.2 Increase to MTOW

At this point in the iterative process the Systems team who oversaw the initial sizing of the UAV performed a major redesign of the battery system. This caused the payload to increase from 28.4 to 51.5 kg . Using procedure 3 and the updated MTOW parameter a new wing planform was found. This caused S_{ref} to increase significantly to 0.67 m^2 .

Though the wing now allowed the UAV to fly slow enough for the camera to pinpoint the migrants' location it was decided to try and reduce the stall speed still further. The lower the stall

speed the less battery power is required for the four vertical propellers to accelerate the UAV up to this speed. Therefore, reducing the stall speed makes the UAV lighter since less battery mass is required. The following section explains how the stall speed was made lower using a multi-aerofoil design.

5.3 Multi-aerofoil design

When analysing the list of aerofoils found from Airfoil Tools the criteria for selection was optimising L/D . However, to reduce the stall speed (with the aim of reducing battery mass) the key indicator is the 2D c_{lmax} . It was realised that for the low Re number towards the tip of the wing the NACA 6412 aerofoil had a 5% higher c_{lmax} than the 4412 aerofoil. Without changing the planform parameters a test was done whereby the aerofoil of the wing was blended from 4412 at the root to 6412 at the tip. The cost of this higher c_{lmax} aerofoil was expected to be a higher drag at the cruise due to it having increased camber. To analyse the increase in drag, Xflr5 was run with the iteration of section 5.2's planform parameters except for the blend of aerofoils. The cruise could be achieved at a lower α however the drag at the cruise was still higher at 19.9 N. To assess how much the stall speed was reduced by with this multi-aerofoil design, procedure 3 was run. In this procedure V_s is fixed while the initial choice of C_{Lmax} is converged to the actual C_{Lmax} . A modification of this method was used such that V_s was allowed to vary until the initial choice of V_s had converged to the value given by a rearrangement of equation 4. To model the local c_{lmax} along the span a blend of the two aerofoil's c_{lmax} was performed as shown in figure 3 .

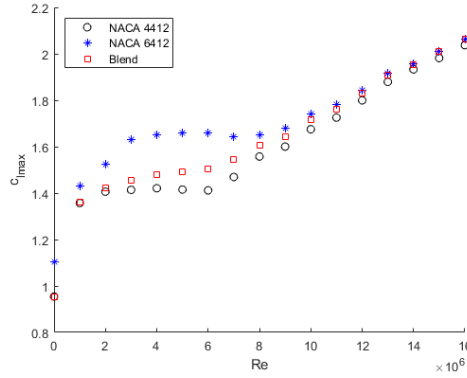


Figure 3: c_{lmax} blend between 4412 and 6412 aerofoils

It was calculated that the stall speed of this new multi-aerofoil design reduced to $V_s = 28.9$ m/s. Given the small increase in local c_{lmax} with the implementation of this design it was not expected that the stall speed would reduce by much. To assess whether the reduction in stall speed was worth the increase in drag a merit index needed to be established which could account for the trade off between the loss in aerodynamic efficiency (lower L/D) compared to the increase in performance due to the reduction in battery mass. A suitable parameter to maximise from both an Aerodynamic and Control perspective is the range of the UAV because despite the mission profile specifying a 70 km range, the greater the range the better. For a propeller driven aircraft, Breguet's equation for range R is given by

$$R = \frac{\eta}{c} \frac{C_L}{C_D} \ln \frac{W_i}{W_f} \quad (9)$$

where η is the propeller efficiency, c is the power specific fuel consumption and W_i and W_f are the initial and final aircraft weights respectively. Assuming that η and c are constant during the first cruise the merit index to maximise is

$$merit = \frac{C_L}{C_D} \ln \frac{W_i}{W_f} \quad (10)$$

A STAR CCM simulation by Leung found that the zero lift drag coefficient (C_{D0T}) of the UAV was 0.066. Though this seemed high it is likely due to the small reference area of the wing compared to the large-diameter fuselage. However, for the sake of the merit index it is only the magnitude of the drag that is important for scaling the aerodynamic efficiency relative to the reduction in mass.

Subtracting the zero lift drag coefficient of the wing (C_{D0}) from C_{D0T} and adding C_D of the cruise for both the 4412 case and the blended wing the total UAV drag coefficient (C_{DT}) at the cruise was determined. These were divided from the C_L at cruise to give the values in table 4.

To calculate the weight ratio, the difference between W_i and W_f was set at the fuel mass used up in the cruise of 1.43 kg. W_f of the 4412 wing was 51.5 kg while W_i and W_f for the blended aerofoil were both reduced by the decrease in battery mass required due to the lower stall speed. This was calculated by Ahlner [13] who found that if the stall speed was reduced to 28.9 m/s then the battery mass required to transition would decrease by 6.2 g. This small reduction in mass is due to the fact that reducing the stall speed does not significantly reduce the time required for the vertical propellers to be spinning.

Table 4: Merit index of 4412 aerofoil compared to blended aerofoil

	C_L/C_D	$\ln \frac{W_i}{W_f}$	<i>merit</i>
NACA 4412	5.0902	0.0282	0.1433
NACA 4412-6412	5.0676	0.0282	0.1427

Table 4 shows that the aerodynamic penalty of the blended aerofoil outweighed the benefit of the reduction in battery mass. At this point it was considered whether other aerofoils besides the NACA 4, 5 and 6 families should be researched to try and find an even higher c_{lmax} . However, this was abandoned because the Propulsion team found that battery mass could not be reduced regardless of any reductions to the stall speed. This was because the battery mass was constrained by the UAV's power requirements in the hover as opposed to the energy requirement of the transition.

5.4 Increase to wing thickness

A request from the Structures team was put forward asking that the wing thickness be increased to allow for the vertical propeller pylons to integrate better into the wing. Since the 4412 aerofoil was already optimised for the current planform, the aerofoil coordinates were simply inflated which increased the thickness to chord ratio giving a 4413 profile. By increasing the thickness without changing the magnitude of the camber nor the camber position the wing did not deviate too significantly from the optimum wing profile.

5.5 Correction to C_{Lmax}

At this point in the iterative design of the wing Leung had performed a STAR CCM analysis on the wing at α near the stall. Leung found that C_{Lmax} of the wing was 1.25 instead of 1.45 given by procedure 2. To better understand the C_{Lmax} of the wing a third method was proposed - an empirical equation. A versatile empirical relation was needed to account for the complex nature of the wing planform which involved both taper and twist. Hence ESDU 89034 [14] was used which states that

$$C_{Lmax} = C_{Lm}/\mu_p + \Delta C_{LM} + \Delta C_{LT} \quad (11)$$

for an unswept wing with twist. C_{Lm} is the maximum lift coefficient for the aerofoil section at the peak value of the normalised lift coefficient c_l/C_L . ΔC_{LM} is the correction factor for Mach number and the shape of the leading edge while ΔC_{LT} corrects for the twist. Using equation 11 C_{Lmax} was found to be 1.19. This provided reason to trust the STAR CCM result of $C_{Lmax} = 1.25$. One reason for AVL over-predicting C_{Lmax} is that it assumes the air flow remains tangential to the wing even at large α . This assumption is valid for small angles of attack but for high α AVL does not model the characteristic separation effect that occurs. To account for this an additional step was added to procedure 2 such that C_{Lmax} was multiplied by a factor of 1.25/1.45=0.86. At this late stage in the design process, there was not time for the FATs to respond to a large change in the wing planform. Hence it was agreed that the iteration of section 5.4 would be used except with an additional increase to the wing span. Due to packing constraints on how small the UAV needs to be when stored the wing span could not be increased by much. It was decided that a 0.1 m increase on each end would be suitable to help alleviate the reduction in C_{Lmax} that had initially been expected from the wing. Hence b increased to 2.2 m and λ reduced to 4.5. This update to the wing effectively constituted an addition of wingtips to the existing design.

Although procedure 2 over-predicted C_{Lmax} , this did not invalidate the horizontal position along the wing of stall. Indeed while looking at this procedure in greater detail it was realised that the stall position of the updated wing (with θ still equal to -2.5°) was at 53.9% of the way along the wing. With the ailerons starting at the 40% position this presented a problem in that, in the event of a stall, loss of roll control might ensue. Hence the variation of twist on stall position was assessed. However, using more washout also has the effect of increasing the drag due to there being a greater area presented to the free stream at any single angle of attack. Figure 4 plots the variation of wing drag at the cruise with stall position as the magnitude of twist was increased. It shows that increasing the washout does move the stall position inboard but at the cost of increased drag. Not only is large washout aerodynamically inefficient, but it is also difficult to manufacture. A compromise was made such that wing had a new twist of $\theta = -4^\circ$. The Control team were able to change the size of the ailerons so they were positioned outboard of 45% along the semispan. Hence when stall occurs only a small portion of the aileron would be affected since with $\theta = -4^\circ$ stall occurs 48.7% of the way along one wing. The final wing parameters are given in table 5.

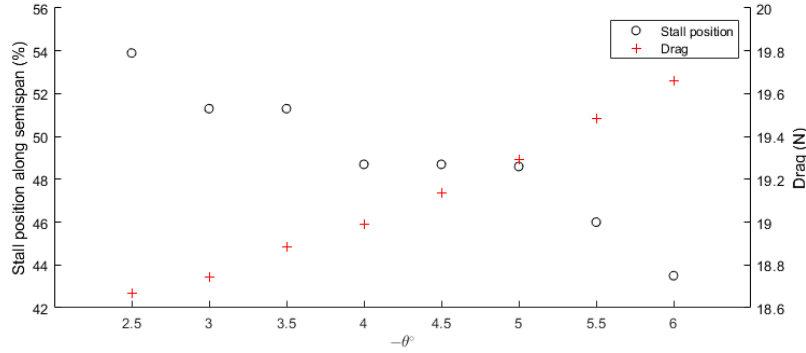


Figure 4: Stall position and wing drag vs twist

Table 5: Final wing parameters

Parameter	Value
Aerofoil	NACA 4413
AR	6.83
c_r (m)	0.444
c_t (m)	0.2
S_{ref} (m ²)	0.709
b (m)	2.2
λ	0.45
θ (°)	-4

6 Wing-propeller interaction using VSPAERO

Given the close proximity of the vertical propellers to the wing, the propeller-wing interaction cannot be neglected. To simulate the effect of this interaction OpenVSP software was used to create a model of the UAV. Then the VSPAERO simulation software in-built within the OpenVSP environment was used to run an inviscid vortex lattice analysis on the UAV in the transition phase of flight. This is where the vertical propellers are spinning while the UAV is simultaneously moving forward. The two methods that VSPAERO can use to model the propellers are as actuator disks or as rotating blades.

In a recent paper by Sheridan [15], the accuracy of VSPAERO simulations was compared to a higher-fidelity RANS simulation. It was shown that the rotating blades method is more accurate than using actuator disks because it captures more of the physics of the flow scenario. The downside is that it requires more computing time. The rotating blades results of this paper were successfully reproduced in VSPAERO to confirm that a thorough understanding of the intricacies of the software had been reached.

The next step in validating the accuracy of VSPAERO simulation was to test a single vertical propeller in the hover condition (0 free stream velocity). The chord and twist distribution of the vertical propeller defined by the Propulsion team were inputted into the OpenVSP software. At an rpm of 6,500 the thrust was 165 N. The Propulsion team had designed the propeller using blade element momentum theory. Using this theory they predicted that one propeller under the same conditions would produce 165 N of thrust. These two values are close to within 10% difference suggesting that the VSPAERO simulation was modelling the rotating blades accurately. However, it was not expected that the thrust would be exactly the same because BEM and vortex lattice models take different approaches to predicting the force from a propeller. For example, vortex lattice models do not account for viscous effects such as wake separation and stall of certain regions of the blade. However, being an unsteady method, VLM does account for the interaction of the shed vorticity between the blades which BEM cannot model given it assumes a steady state.

To mount the four vertical propellers to the UAV, they were designed to be attached to pylons which are secured into the leading and trailing edges of the wing structure. Ideally, an analysis of the effect of propeller wake should account for the pylon's structure. However, this was difficult to conduct in VSPAERO. The reason being that, just as AVL models the 3D wing's camber but not its thickness, VSPAERO simplifies a 3D body to two planes to account for the width and height of the body. If this method had been used to try and model the pylons it would have been difficult to assess whether the results were accurate for the full 3D shape or if the 2D planar representation was insufficient. Hence the four 2-bladed propellers were situated around the main wing and the pylons were not modelled.

Multiple simulations were run for different rpm and forward velocities. The test case which was analysed in detail was the transition phase where the wing produces a quarter of the required lift while the propellers produce approximately three quarters between them. In free stream flow of $V = 17$ m/s with the wing set at $\alpha = 9.8^\circ$ and with no propellers the wing produces 139.5 N of lift. A single vertical propeller in a hover was found to produce 100.1 N of thrust at 5,000 rpm. Hence simulations were run with the four vertical propellers operating at 5,000 rpm and the wing set at $\alpha = 9.8^\circ$ in a 17 m/s flow. The simulation parameters are outlined in table 6.

Table 6: VSPAERO simulation parameters

Parameter	Value
Batch calculation	On
Second order Karman-Tsien Mach Correction	On
Wake Nodes	128
Far Field Distance	50
Propeller	
Auto Time Step	On
Num Revs	5
From Steady State	On

The Auto Time Step flag automatically adjusts the time step such that the simulation is updated when the propellers rotate by 18° and the number of time steps is calculated to satisfy the number of revolutions (Num Revs) specified. The From Steady State flag requests that VSPAERO first model the flow around the wing before rotating the propeller blades. This is required because, over the small time periods of a few propeller revolutions the flow does not have time to fully develop over the wing. After trialling multiple choices of for the number of revolutions a final value of 5 was set in order for the angle of the propeller wakes to have converged to a constant direction after having turned the free stream flow.

Three different propeller positions were analysed: in the plane of the wing and either 6 cm above or below the plane of the wing. Figure 5 shows the thrust produced by a single front or back propeller in each of the three cases over the last 2.5 propeller revolutions. Only the data for the left hand front and back propellers are shown since the flow is symmetrical about the fuselage and the thrust data showed this symmetry. The thrust of all the propellers showed considerable oscillation in magnitude over the small time periods that were tested. This was due to the forward swinging blade of each propeller creating more lift when perpendicular to the free stream and both blades producing less lift when parallel to the free stream. This occurred with a time period of half one revolution. Moreover, in all three cases, the back propellers oscillated about a mean thrust value

that was lower than the the front propellers. This was due to the front propellers' wakes blanketing the rear propellers and reducing the lifting ability of the blades. Figure 5 shows that, regardless of the propellers' vertical position, the thrust produced by each propeller was similar. This means that the wing's position relative to the propeller did not have any significant effect on altering the propeller thrust produced. Both the front and back propellers produced 40 and 18 N more thrust respectively than the hover value of 100.1 N.

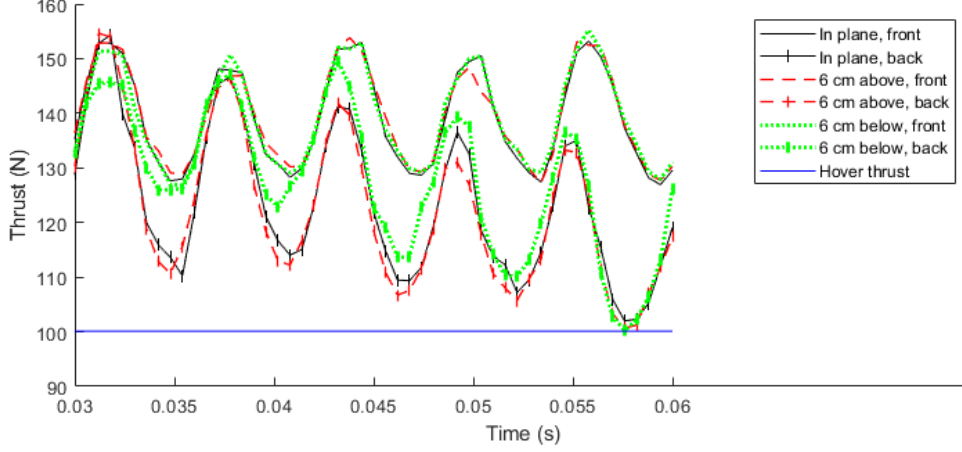


Figure 5: Propeller thrust over last 2.5 revolutions

The lift from the wing in the presence of the trailing wakes of the propellers showed significant temporal oscillations. Even after the unsteadiness of the simulation had subsided there was still variation in lift about a mean value. This is evident in figure 6 which shows the wing lift for the three cases. The dashed lines are averages over the last 0.01 seconds of the simulation.

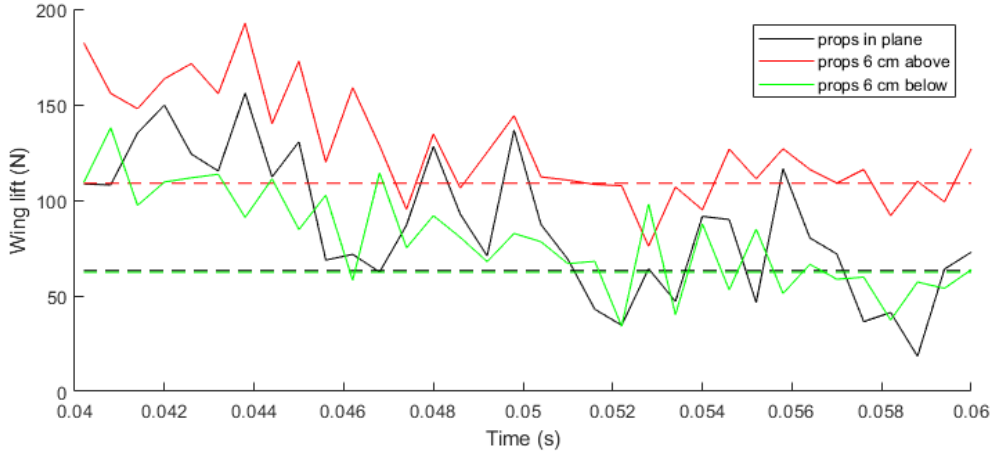


Figure 6: Wing lift vs time for the three propeller positions

Figure 6 clearly shows one of the downsides of the vortex lattice method that VSPAERO uses in that it does not account for the inertia of the air flow. It is unrealistic for the wing's lift to change by over 100 N in a period of 0.002 seconds (propellers in plane from 0.05 to 0.052 s). However, the average of the data more to accurate in modelling the true lift produced by the wing. The average lift (L_{ave}) produced by the wing with the high positioning of the propellers was 108.8 N. L_{ave} for the in plane and low positioning of the propellers was 63.2 and 62.3 N respectively. These two latter propeller positions cause the wing to produce less lift in comparison to the no-propeller wing lift of 139.5 N. This is likely to be due to the propellers' wakes' accelerated flow all being directed below the wing creating lower pressure in this region which reduces the pressure differential above and below the wing surface. More evidence for the validity of this hypothesis is figure 7 which shows some of the trailing wakes from the front propellers still flowing over the top surface of the wing due to the high positioning of the propellers.

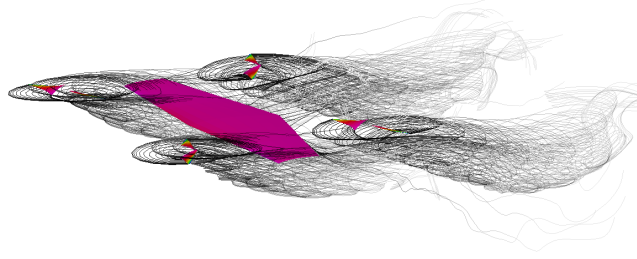


Figure 7: Trailing wakes from propellers 6 cm above wing

Of the three vertical positions of the propellers, mounting them above the wing is most desirable because of the greater lift from the wing that is created. One downside of this choice which VSPAERO has not modelled is the effect of stalled blades. Particularly at high forward velocities when the propellers will be operating far from their optimum of being in the hover state blade stall could occur causing separation. This wake of high vorticity has been shown to most likely impinge on the wing in the case of high-mounted propellers. Without undergoing detailed RANS simulations it is, however, difficult to model the effect of stalled blades. On the other hand another advantage of positioning the propellers above the wing is that this means they are attached above the wing pylons. This results in them seeing a clean inflow. If they were placed below the pylons then their efficacy would be reduced due to the pylon structure restricting the mass flow rate available to them.

7 Conclusions

To conclude, an optimal aerofoil and wing design has been chosen which satisfies the constraints imposed by the mission profile as well as the requirements of the other design teams. After optimising the aerofoil selection and wing planform under aerodynamic considerations alone the area of the wing was increased to provide a much lower stall speed.

A merit index was introduced to help quantify the trade off between how different variables should be balanced when there were competing interests between FATs. However this merit index was sometimes not able to be utilised because often the requirements of one FAT were vital to the success of the mission and hence took precedence over the competing FAT's desires. In future preliminary design of this aircraft the codes and analysis procedures used by each FAT should be integrated into a large, multi-disciplinary dashboard where all the design parameters can be changed and their effects immediately analysed to achieve an optimum design. Given the short span of this project, such an automated integration was not achieved. However, a significant level of interaction did occur - but this involved incremental improvements in the form of iterations to the wing as explained in the report.

The analysis of the optimum placement of the vertical propellers and their interaction with the wing was analysed. It was found that there are no simple and computationally-inexpensive methods of accurately modelling the propeller wake interaction. Though insight has been gained through use of VSPAERO, it was limited to analysing the 4 propellers in conjunction with a camber-representation of the wing. The effect of the vertical placement of the propellers was analysed. It was found that thrust did not depend on vertical position but wing lift did. Furthermore, regardless of the vertical position, the rear propellers all had a reduced thrust output due to the blanketing effect. In further studies on this UAV, a full RANS simulation needs to be set up to account for the air flow around the pylons. It could be deemed necessary to conduct a DNS simulation too to be able to accurately capture the effect of temporal oscillations on the pylons to aid the Structures team in designing for fatigue.

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Appendix

The code to run AVL from the command line is executed by running myMain.m which calls avl_give_cd.m

myMain.m is defined below

```

clear
clc
close all

cd 'C:\Users\danny\OneDrive\1-_-GDP\Athena_Vortex_Lattice\code_avl'

load('read_xfoil_cd_data_vars', 'xfoil0')
avl_give_cd_input.xfoil0 = xfoil0;

lambda_all = [1:-0.05:0.2];
twist_all = [0:-0.25:-4];

num_vars = 4;
Cd_store = zeros(length(twist_all), length(lambda_all), num_vars);
% down in twist, along in lambda
% the 3D part of Cd_store goes through Cdi, Cdv and Cd (Cd=Cdi+Cdv)

for lambda_index = [1:length(lambda_all)]
    for twist_index = [1:length(twist_all)]

        avl_give_cd_input.lambda = lambda_all(lambda_index);
        avl_give_cd_input.twist = twist_all(twist_index);

        [output] = avl_give_cd(avl_give_cd_input);
        if output.status=="bad"
            break
        end

    dim_3_index = 1;

```

```

Cd_store(twist_index , lambda_index , dim_3_index) = output.Cdi;

dim_3_index=dim_3_index+1;
Cd_store(twist_index , lambda_index , dim_3_index) = output.Cdv;

dim_3_index=dim_3_index+1;
Cd_store(twist_index , lambda_index , dim_3_index) = output.Cd;

dim_3_index=dim_3_index+1;
Cd_store(twist_index , lambda_index , dim_3_index) = output.e;

fprintf( '%u | %u of %u | %u is complete \n' , ...
twist_index , lambda_index , length(twist_all) , ...
length(lambda_all))

end

if output.status=="bad"
fprintf( '\n error . at lambda=%f , twist=%f we tried ...
.....interpolating outside the given range \n' , ...
avl_give_cd_input.lambda , ...
avl_give_cd_input.twist)
break
end
end

myMain_output.Cd_store = Cd_store;
myMain_output.lambda_all = lambda_all;
myMain_output.twist_all = twist_all;
save( 'myMain_vars' , 'myMain_output')

The function avl_give_cd.m was implemented as follows

function [output] = avl_give_cd(avl_give_cd_input)

output.status = "ok";

cd 'C:\Users\danny\OneDrive\1_GDP\Athena_Vortex_Lattice\code_avl'

% make myPlane.avl
WS = 1058.6;
MIOW = 28.4; % kg
S_ref = MIOW * 9.81/WS;
AR = 7;
b = sqrt(AR*S_ref); % span
rho = 1.21; % at 400 ft
u_cruise=58.3; % m/s
cl_des_3D = WS/(0.5*rho*u_cruise^2);
lambda = avl_give_cd_input.lambda; % taper ratio
twist = avl_give_cd_input.twist; %degree
c_root = 2*S_ref/(b*(1+lambda));
c_tip = lambda*c_root;

% calculate MAC
syms y
x = c_root + (c_tip-c_root)*2/b*y;
integral_value = int(x^2, y, 0, b/2);
Cref = double(2/S_ref * integral_value);

Nchord = 10;

```

```

Cspace = 1;
Nspan = 8;
Sspace = 1;

setting_angle = 0;

root_x_le = c_root - c_tip;

naca_number = 2410;
semispan = b/2;
tip_angle = twist;

fid=fopen('myPlane.avl','w+'); % creates the input file

fprintf(fid,'myPlane\n');
fprintf(fid,'0.0-----!Mach\n');
fprintf(fid,'0-----0.0-----!iYsym--iZsym--Zsym\n');
fprintf(fid,'%f%f%f-----!Sref--Cref--Bref--reference_area,_chord,_spa
Cref, b);
fprintf(fid,'3.00--0.0--0.5-----!Xref--Yref--Zref--moment_reference_location
fprintf(fid,'0-----!CDp\n');
fprintf(fid,'#\n');
fprintf(fid,'#-----\n');
fprintf(fid,'#\n');
fprintf(fid,'SURFACE\n');
fprintf(fid,'Wing\n');
fprintf(fid,'%f%f%f%f%!Nchord--Cspace--Nspan--Sspace\n', Nchord,
Cspace, Nspan, Sspace);
fprintf(fid,'#_reflect_image_wing_about_y=0_plane\n');
fprintf(fid,'YDUPLICATE\n');
fprintf(fid,'0\n');
fprintf(fid,'#_twist_angle_bias_for_whole_surface\n');
fprintf(fid,'ANGLE\n');
fprintf(fid,'%f\n', setting_angle);
fprintf(fid,'SCALE\n');
fprintf(fid,'--1.0--1.0--1.0\n');
fprintf(fid,'#_x,y,z_bias_for_whole_surface\n');
fprintf(fid,'TRANSLATE\n');
fprintf(fid,'0_0_0\n');
fprintf(fid,'#\n');
fprintf(fid,'#-----\n');
fprintf(fid,'#\n');
fprintf(fid,'#Xle-----Yle-----Zle-----chord-----angle\n');
fprintf(fid,'SECTION\n');
fprintf(fid,'0-----0.0-----0.000\n', c_root);
fprintf(fid,'NACA\n');
fprintf(fid,'%d\n', naca_number);
fprintf(fid,'\n');
fprintf(fid,'#-----\n');
fprintf(fid,'#Xle-----Yle-----Zle-----chord-----angle\n');
fprintf(fid,'SECTION\n');
fprintf(fid,'-----%f-----0-----%f-----%f\n', root_x_le, semispan, c_t
fprintf(fid,'NACA\n');
fprintf(fid,'%d\n', naca_number);
fprintf(fid,'\n');

fclose(fid);

%%%%%%%%%%

```

```

% make my_command_show_fs.txt
fid=fopen('my_command_show_fs.txt','w+'); % creates the input file

fprintf(fid,'LOAD_code_avl/myPlane.avl_\n');
fprintf(fid,'OPER_\n');
fprintf(fid,'Cl_\n');
fprintf(fid,'D_1.2200_\n');
fprintf(fid,'V%f_\n', u_cruise);
fprintf(fid,'\n');
fprintf(fid,'a_\n');
fprintf(fid,'C%f_\n', cl_des_3D);
fprintf(fid,'L_\n');
fprintf(fid,'X_\n');
fprintf(fid,'FS_\n');
fprintf(fid,'FS_strip_forces.txt_\n');
fprintf(fid,'FT_\n');
fprintf(fid,'FT_total_forces.txt_\n');
fprintf(fid,'\n');
fprintf(fid,'quit_\n');

fclose(fid);

cd .. % go up one folder to where we can run avl.exe
[~,~]=dos('del_FT_total_forces.txt'); % delete file otherwise matlab will complain wh
[~,~]=dos('del_FS_strip_forces.txt');

[~,~]=dos('.\avl.exe<code_avl/my_command_show_fs.txt'); % runs avl, outputting text

%% Reading Total forces
fid=fopen('FT_total_forces.txt','r+');

output.alpha=ReadFile(fid,'Alpha'); % AoA
output.Cl=ReadFile(fid,'CLtot'); % Cl
output.Cdi=ReadFile(fid,'CDff'); % lift=induced Cd
output.e=ReadFile(fid,'e'); % oswalds

fclose(fid);

%% Reading Strip forces
fid=fopen('FS_strip_forces.txt','r+');

ReadFile(fid,'cm_c/4');
data=fscanf(fid,'%f',[15,Nspan]); % scans full data table
fclose(fid);

% data

% Processing of strip forces
span_output.y=data(3,:); % y-positions
span_output.c=data(5,:); % chords
area = data(6,:);
ccl=data(7,:); % cl * section chord
w.cbar = Cref;
span_output.cl_cref=ccl/w.cbar; % cl, using MAC as reference chord

```

```

span_output.cl_c=ccl./span_output.c; % cl, using section chord as reference chord
% I think this is the sectional lift coefficient (lift coefficient per unit span)
span_output.alphai=data(8,:); % induced AoAs
span_output.cdi=data(11,:); % induced Cd

% contents of conv9_outproc.m
bool.dhdrfudge=0; % due to dihedral, the lift and drag at the root spike up.
% if this is 1, rootmost point is neglected when
% interpolating.

fi=bool.dhdrfudge+1; % 1 if no fudge, 2 if fudge (skips rootmost point in distribution)

w.s = semispan;
w.S = S_ref;
span_output.ss=span_output.y/w.s; % spanwise stations - 0 at root, 1 at tip
% w.s is the semispan defined in conv4
% span_output.y is y positions of strips
% so span_output.ss is a spanwise station array but not as fine as sss
sss=0:0.001:1; % finer spanwise station array
cls=interp1(span_output.ss(fi:end),span_output.cl_cref(fi:end),sss,'pchip'); % interpolation
% btw: span_output.cl_cref=ccl/w.cbar; % cl, using MAC as reference chord
cdi=interp1(span_output.ss(fi:end),span_output.cdi(fi:end),sss,'pchip'); % interpolation

w.ss = [0,1];
w.i0=setting_angle;
w.twist=[0,twist];
w.i=w.i0+w.twist;
is=interp1(w.ss,w.i,sss,'linear'); % incidences along span

as=is+output.alpha - interp1(span_output.ss(fi:end),span_output.alphai(fi:end),sss,'pchip');
output.as=as;

%% Viscous drag estimation
w.c = [c_root, c_tip];
cs=interp1(w.ss,w.c,sss,'linear');
% me: these are the chords

% Don't use atmosisa for now
% [T,~,~,rho]=atmosisa(fc.alt);
T = 288;

mu=1.716e-5*(T/273)^1.5*(273+111)/(T+111); % viscosity calculation
fc.M = 0; % Mach number
% Res=rho*fc.M*sqrt(1.4*287*T)*cs/mu;
Res=rho*u_cruise*cs/mu;

% use xfoil data which gives cd as a function of reynolds and alpha to: find cd for each
% [xfoil0] = read_xfoil_cd_data();
xfoil0 = avl_give_cd_input.xfoil0;
% xfoil.re has same rows. reynolds increases going across.
% xfoil.a has same columns. alpha increases going down.
interpolation_range = [xfoil0.re(1,1), xfoil0.re(1,end);
xfoil0.a(1,1), xfoil0.a(end,1)];

if not(all(Res >= interpolation_range(1,1)) & all(Res <= interpolation_range(1,2)) & all(Res > 0))
    output.status="bad";
return

```

```

end

cd0=interp2(xfoil0.re,xfoil0.a,xfoil0.cd,Res,as);

cdv = cd0;
Kdv = trapz(sss*w.s, cdv.*cs)*2; % integrates viscous drag along the wing
output.Cdv = Kdv/w.S;

output.Cd=output.Cdi+output.Cdv;

% other outputs
output.Cl = cl-des_3D;
output.AR = AR;

fprintf('cl-des_3D: %f\n', cl-des_3D)

cd code_avl

% write wing planform parameters to xml file which can be opened in xflr5
if 0
    write_xml_input.naca_number = naca_number;
    V_s = 999;
    write_xml_input.V_s = V_s;
    write_xml_input.semispan = semispan;
    write_xml_input.c_root = c_root;
    write_xml_input.root_x_le = root_x_le;
    write_xml_input.c_tip = c_tip;
    write_xml_input.tip_angle = tip_angle;
    write_xml_input.Nchord = Nchord;
    write_xml_input.Nspan = Nspan;

    write_xml_input.filename = 'look_at_cdi_cdv';

    write_xml(write_xml_input)
end

end

```